



Introduction to

Data Integration

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Keywords:

Data Integration, Registered data, domain, domain name system, web, distributed database, database schema, Heterogeneities, Model Heterogeneities, Data model, Synonyms, Homonyms, Attribute, Entity

Introduction to **Data Integration**

In this lecture:



- Part 1: **Example of a Data Integration Problem**
- Part 2: Challenges of Data Integration
- Part 3: Practice
- Part 4: Discussion

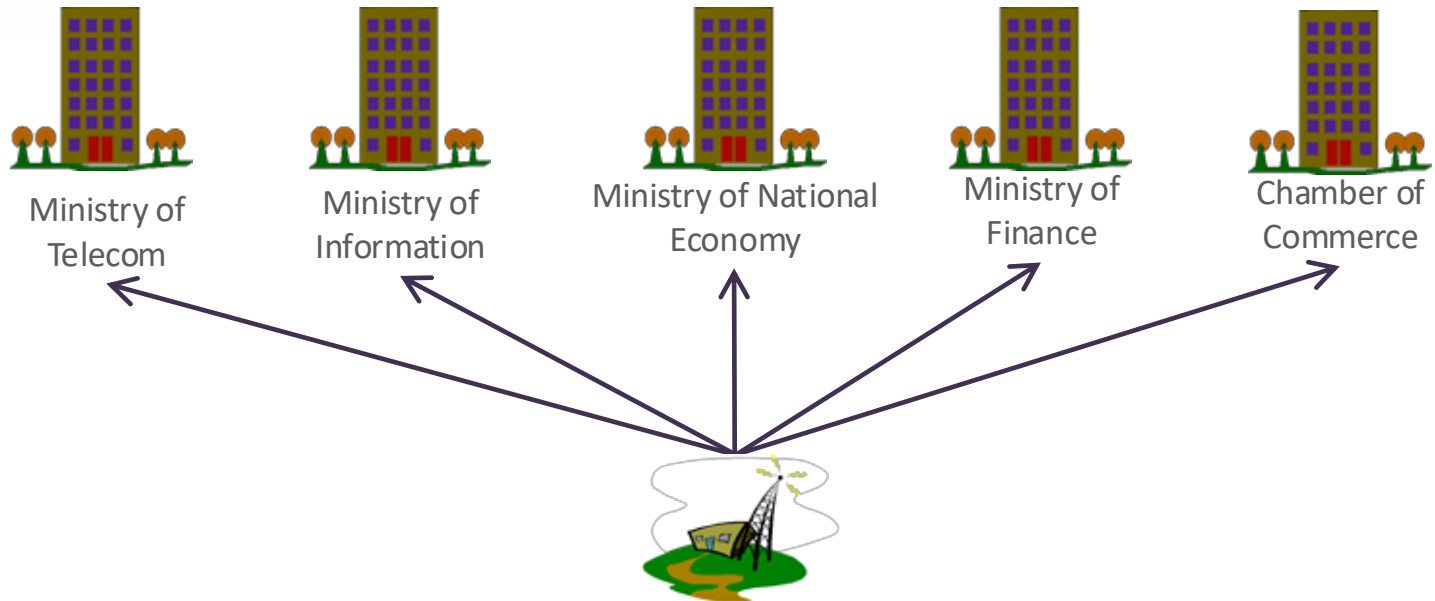
Keywords:

Interoperability, Data Integration, Distributed Databases, Schema Heterogeneity, Name Heterogeneity, Semantic Heterogeneity, Structural Heterogeneity, Model Heterogeneity, Data Model, Relational Model, RDF Model, Data Web, Global Identifier, Schema Mapping, Data Consistency

Example from the Government Domain

Consider all interactions with government agencies in order to register a new business.

Example: Establishing a new Radio Station in Palestine, which involves 5 agencies.

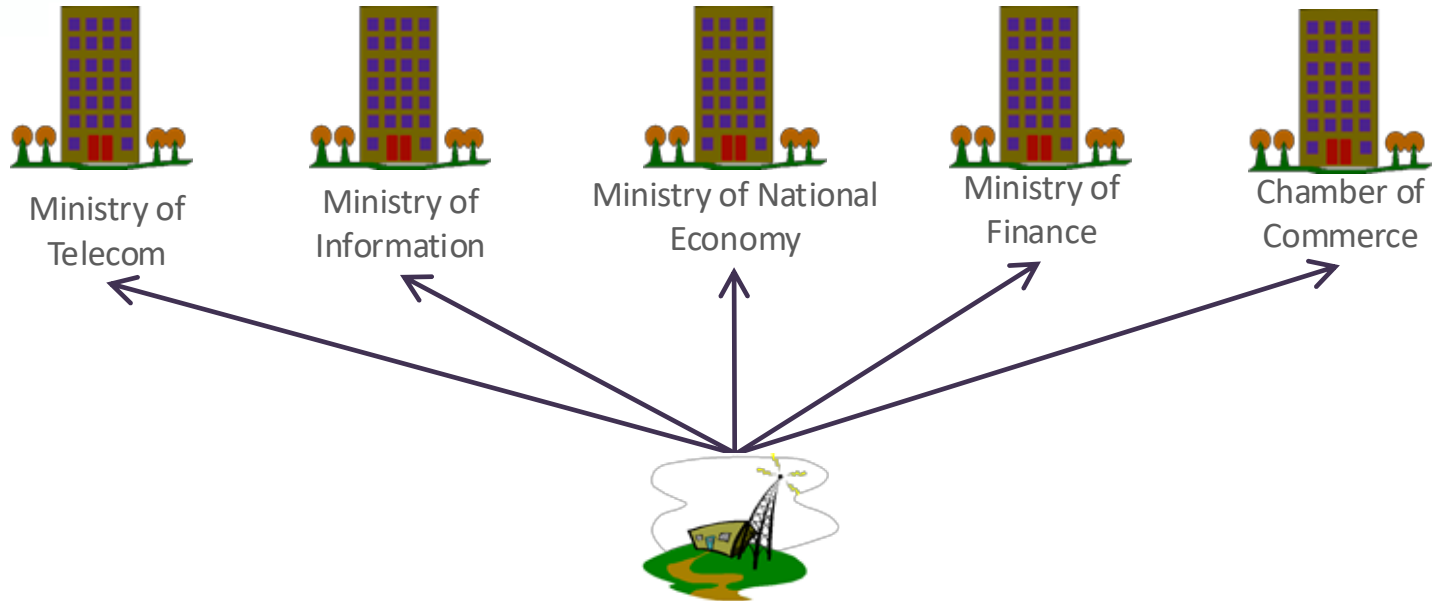


Example from the Government Domain

When your business evolves or changes.

Example: Changing the address of the radio station.

- Address must be changed in 5 different databases.



Example from the Government Domain

Consider the data registered about the same radio station in the databases of different ministries and governmental agencies:

Agency 1	ID	Name	Type	City
	R2563I	Radio Al-Amal	Radio Station	Ramallah
Agency 2	B_ID	Business Name	Activity Type	Province
	LM1847	Al-Amal Broadcast	Radio Broadcasting	Ramallah and Bireh
Agency 3	ID	Company Name	Company Type	Location
	182NS3	'Broadcast Al-Amal	Broadcasting Station	Al-Balu'

Example from the Government Domain

From this example one can point out to some challenges in Data Integration:

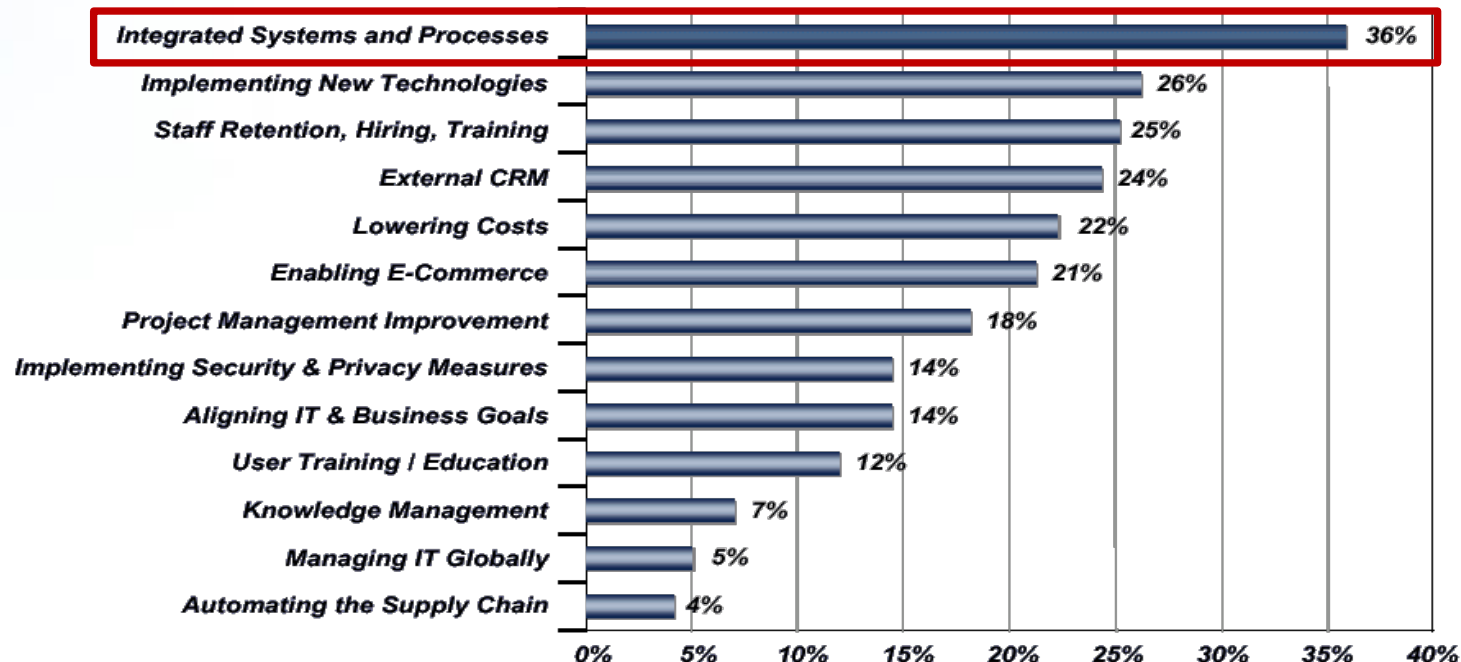
- No agreed upon **naming** (name, business name, company name)
- No agreed upon **meaning** (Does 'Activity Type' mean exactly the same as 'Company Type'?)
- Different **Registered Data**: Radio Al-Amal, Al-Amal Broadcast,

Agency 1	ID	Name	Type	City
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Agency 2	B_ID	Business Name	Activity Type	Province
	LM1847	Al-Amal Broadcast	Radio Broadcasting	Ramallah and Bireh
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	182NS3	Broadcast Al-Amal	Broadcasting Station	Al-Balu'

...

The Problem is in All Domains

Top IT Spending Priorities¹

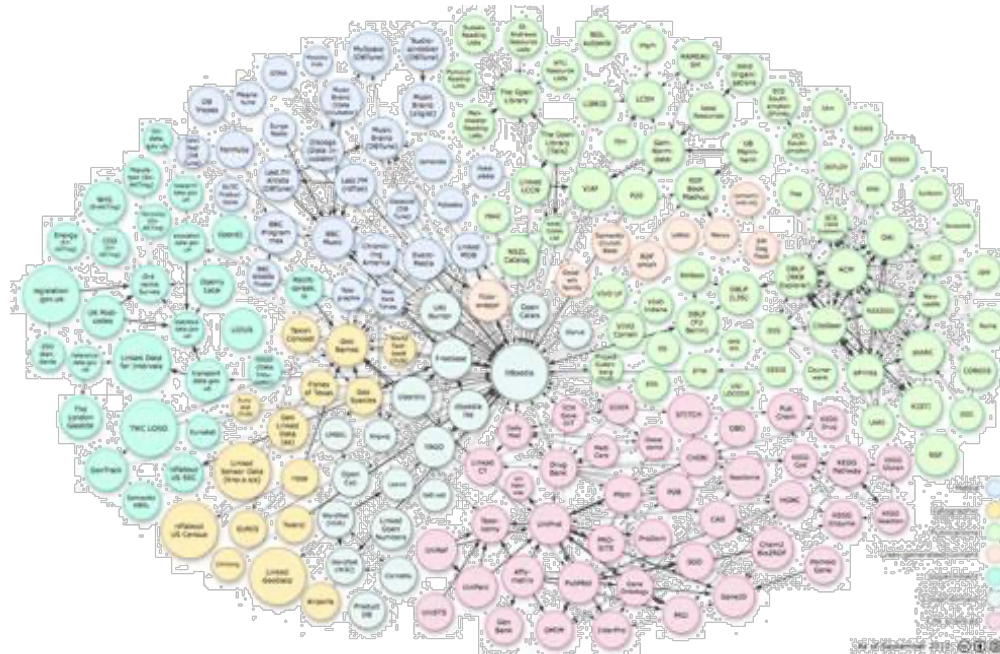


¹CIO Magazine Survey, February

2002
33% of firms surveyed have EAI projects (Forrester, March 2002 Business Technographics benchmark)

The Problem is in All Domains

- Problem is now even more challenging with the Web.
- The Data Web envisions the web as a global world-wide database.
- This means that one can query distributed multiple databases on the web as if he/she is querying a local database.



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Challenges of Data Integration:

Heterogeneities in Database Schemas

One can distinguish between several heterogeneities between different schemas:

- **Name** Heterogeneities (difference in used vocabulary).
- **Meaning** Heterogeneities (different meaning/semantics for the same attribute in two schemas).
- Heterogeneities in the **structure and type**.
- Heterogeneities in the **rules and constraints**.
- **Data Model** Heterogeneities.

Name and Meaning Heterogeneities

Synonyms – different names for the same concepts

- employee, clerk
- exam, course
- code, num

Homonyms – same name for different concepts (different meanings)

- City as *City of birth* in one schema,
- City as *City of Residence* in another schema

Salary: Net Salary

Salary: Gross Salary

Homonyms

Section

Division

A specialized
division of a large
organization

Synonyms

Heterogeneities in Structure and Type

Source: Carlo Batini

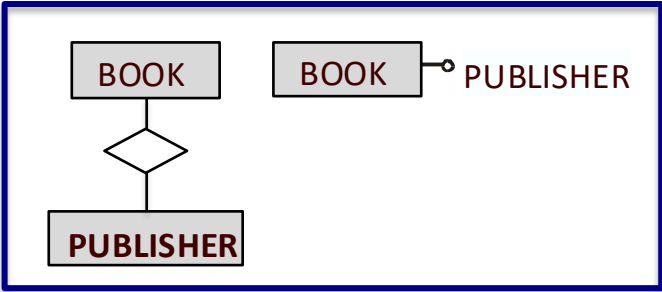
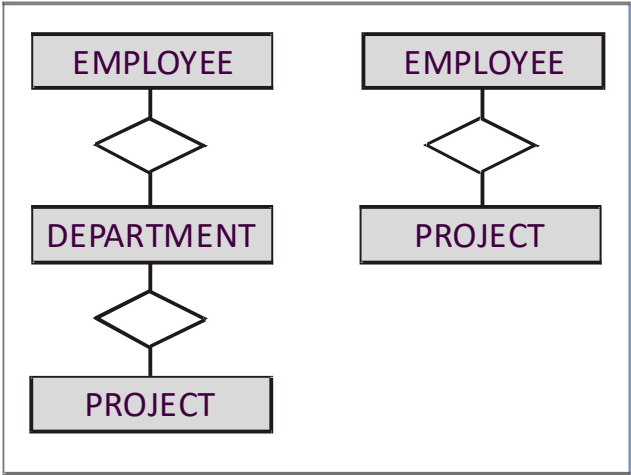
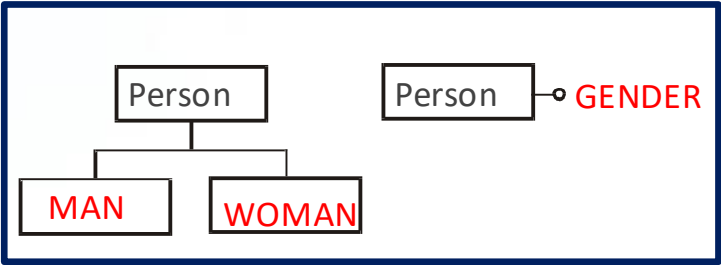
The same concepts are represented with different conceptual structures in two schemas:

- **Attribute** in one schema and **derived value** in another schema.
- **Attribute** in one schema and **entity** in another schema.
- **Entity** in one schema and **relationship** in another schema.
- Different **abstraction levels** for the same concept in two schemas:
e.g. two entities with homonym names related by an IS-A hierarchy in two schemas.

Heterogeneities in Structure and Type

Source: Carlo Batini

Examples:



Heterogeneities in Type

Examples:

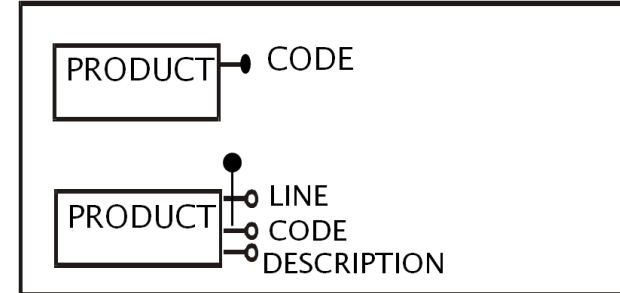
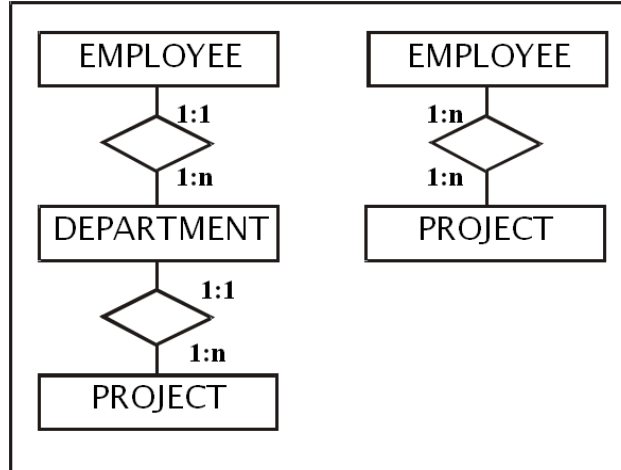
- In a single attribute (e.g., Numeric, Alphanumeric). E.g., the attribute “gender”:
 - Male/Female
 - M/F
 - 0/1
- Year has a four digit domain in one schema and two digit domain in another schema
- Different currencies (Euros, US Dollars, etc.)
- Different measure systems (kilos vs. pounds, centigrade vs. Fahrenheit.)
- Different granularities (grams, kilos, etc.)

Heterogeneities in the rules and constraints

Source: Carlo Batini

Examples:

- Different cardinalities in the same relationships
- Key conflicts



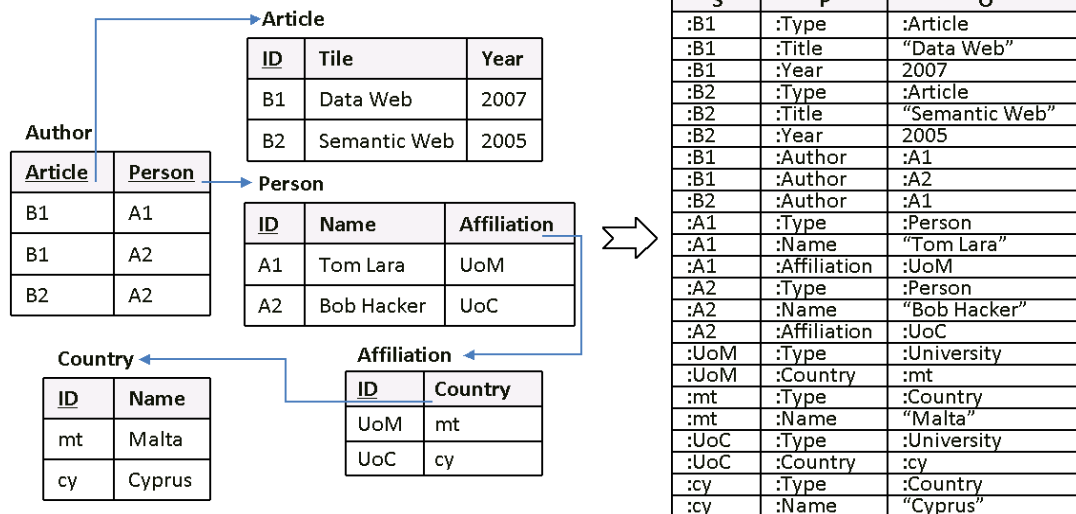
Model Heterogeneities

Model Heterogeneities occurs when different databases adheres to different data models:

- Relational Data Model, XML, RDF, Object-Oriented, OWL, ...

Solution: Reduce Model Heterogeneity by using one data model.

Example: Convert the Relational Model to RDF graph model.



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Open Ideas

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Discussion

Which parties and countries are involved when buy coffee from Tim using your VISA card? What what data integration problems are expected?

What integration problems are expected when you transfer money from you account in Cairo Bank in Egypt to AA Bank in Belgium?

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